Municipal Financial Management System (MFMS)

Course: PAP521S – Programming in Practice (NUST)
Project: Project A
Group: Project A

Group Members

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Project Description

MFMS is a console-based program written in C that helps a municipal finance office manage employees, budgets, suppliers and assets from one menu. It is split into modules (one `.c` and `.h` file each) and keeps its records in memory while the program runs.

System Features

- **Main menu** – a looping menu for all modules; invalid options are rejected.
- **Employee management** – add, display and search employees; salary is calculated as basic + housing + transport + other allowance; employee report.
- **Budget management** – 7 departments; enter allocated budget and expenditure, rename and search departments, list over-budget departments; balance = allocated – expenditure; budget report.

- **Supplier management** – register, display and search suppliers (by ID or partial name); compare two suppliers; e-mail and telephone validation; duplicate e-mail check; supplier report.
- **Asset management** – add, display and search assets (by ID, name, type or department); update asset condition; total asset value; asset report.
- **Reports** – one menu that shows the employee, budget, supplier and asset reports.
- **Input validation** – empty text, invalid numbers, out-of-range values and invalid menu choices are rejected.

Files

`main.c` , `employees.c/.h` , `budget.c/.h` , `suppliers.c/.h` , `assets.c/.h` , `reports.c/.h`

Compilation Instructions

Requires a C compiler such as GCC (on Windows, MinGW-w64). Open a terminal in the project folder and compile all the `.c` files together.

Windows (PowerShell):

```
gcc (Get-ChildItem *.c).FullName -o main
```

Windows (Command Prompt), Linux or macOS:

```
gcc main.c employees.c budget.c suppliers.c assets.c reports.c -o main
```

This creates the executable `main.exe` on Windows, or `main` on Linux/macOS. Compile again after any change to the code.

How to Run the System

Windows (PowerShell):

```
.\main.exe
```

Linux / macOS:

```
./main
```

Enter a number from the main menu (1–6) to open a module: 1 Employee Management, 2 Budget Management, 3 Supplier Management, 4 Asset Management, 5 Reports, 6 Exit. Each module has its own sub-menu with a "back" option that returns to the main menu.

Individual Responsibilities

Member	Responsibility
Wame Griffiths	Main program and main menu (`main.c`)
Ameer Majiet	Asset management (`assets.c` , `assets.h`)
Lasarus Lucas	Part 1: Employee management (`employees.c` , `employees.h`)
Ethan Khembo	GitHub coordination, integration, testing and documentation
Shikongo Gerson	Part 3: Supplier management (`suppliers.c` , `suppliers.h`); header files
Thomas Nikanor	Part 2: Budget management (`budget.c` , `budget.h`)
Werner Maria	Reports module (`reports.c` , `reports.h`)